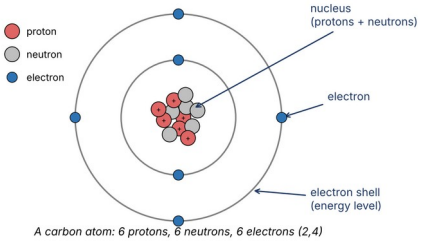
Atomic Structure

AQA GCSE Combined Science: Trilogy · Chemistry Topic 1 · Revision activity 1 · 23 marks · about 25 minutes

Read the key information (5 min), answer the **quick recall** questions without looking back (5 min), then answer the **exam questions** (15 min). Mark your work with the mark scheme on the last page.

Key information

- Everything is made of **atoms**. An atom is the smallest part of an element that can exist.
- An atom has a tiny central **nucleus** containing **protons** and **neutrons**. **Electrons** move around the nucleus in **shells** (energy levels).
- Atoms have **no overall charge** because the number of electrons equals the number of protons.
- **Atomic number** = number of protons. **Mass number** = number of protons + number of neutrons.
- Atoms are very small: the radius of an atom is about **0.1 nm** (1×10^{-10} m). The nucleus is about **1/10 000** of the radius of the atom, but almost all of the mass is in the nucleus.



A carbon atom: 6 protons, 6 neutrons, 6 electrons (2,4)

The three subatomic particles

Particle	Relative charge	Relative mass	Where it is found
proton	+1	1	in the nucleus
neutron	0	1	in the nucleus
electron	−1	very small	in shells around the nucleus

Working out the numbers of particles

The periodic table shows the mass number (top) and atomic number (bottom), e.g. $^{23}_{11}$**Na**

protons = atomic number = 11 **electrons** = protons = 11 **neutrons** = mass number − atomic number = 23 − 11 = 12

Electronic structure

Electrons fill the shells closest to the nucleus first. For the first 20 elements: **1st shell holds up to 2**, **2nd shell up to 8**, **3rd shell up to 8**.

Example: sodium has 11 electrons, so its electronic structure is **2,8,1**.

Section A: Quick recall (10 marks – 1 mark each)

1 What is the relative charge of a proton?

2 What is the relative charge of a neutron?

3 What is the relative charge of an electron?

4 Which two particles are found in the nucleus?

5 Where are electrons found in an atom?

6 What does the atomic number tell you?

7 What does the mass number tell you?

8 Why do atoms have no overall charge?

9 How many electrons can the first shell hold?

10 What is the approximate radius of an atom?

Section B: Exam questions

01 An atom of fluorine has atomic number 9 and mass number 19.

01.1 Give the number of protons, neutrons and electrons in this atom.

[3 marks]

01.2 Give the electronic structure of fluorine.

[1 mark]

02 Copy and complete the table.

Atom	Atomic number	Mass number	Protons	Neutrons	Electrons	Electronic structure
magnesium	12	24				
aluminium	13	27				
potassium	19	39				

[3 marks]

03 Draw a diagram to show the electronic structure of an aluminium atom (atomic number 13).

[2 marks]

04 Explain why almost all of the mass of an atom is in the nucleus.

[2 marks]

05 The radius of an atom is about 0.1 nm. The radius of its nucleus is about 1/10 000 of the radius of the atom. Calculate the radius of the nucleus in nm.

[2 marks]

Mark Scheme – Atomic Structure

(1) = one mark. / separates acceptable alternatives. Words in brackets are not needed for the mark.

Question	Marking guidance	Comments	Mark
A	**1** +1 **2** 0 (no charge) **3** −1 **4** protons and neutrons **5** in shells (energy levels) around the nucleus **6** the number of protons **7** the number of protons + neutrons **8** same number of protons (+) and electrons (–) **9** 2 **10** 0.1 nm (1×10^{-10} m)	1 mark each	10
01.1	protons = 9 (1) neutrons = 19 – 9 = 10 (1) electrons = 9 (1)		3
01.2	2,7 (1)		1
02	magnesium: 12 protons, 12 neutrons, 12 electrons, 2,8,2 (1) aluminium: 13 protons, 14 neutrons, 13 electrons, 2,8,3 (1) potassium: 19 protons, 20 neutrons, 19 electrons, 2,8,8,1 (1)	1 mark for each fully correct row.	3
03	three shells drawn around a nucleus (1) 2 electrons in the first shell, 8 in the second, 3 in the outer shell (1)		2
04	protons and neutrons are in the nucleus and each has a relative mass of 1 (1) electrons have a very small mass (1)		2
05	0.1 ÷ 10 000 (1) = 0.00001 nm (1×10^{-5} nm) (1)		2

Total: 23 marks